FF-C-436f
April 5, 1965

SUPERSEDING
Int. Fed. Spec. FF-C-00436c (GSA-FSS)
December 18, 1964 and
Fed. Spec. FF-C-436c
October 20, 1960

FEDERAL SPECIFICATION

CLIP, PAPER, WIRE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers Gem- and Ideal-pattern types of wire paper clips for general office use.

1.2 Classification.

1.2.1 *Types and sizes.* The clips shall be of the following types and sizes, as specified in the invitation for bids (see 3.2 and 6.1):

Type I—Gem pattern (fig. 1).
Size 1—Large.
Size 2—Small.

Type II—Ideal pattern (fig. 2).
Size 1—Large.
Size 2—Medium.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-P-121—Paper, Bond and Writing, White and Colored.
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

FSC 7510

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** The clips shall be made of steel wire with a tin-coated finish (see 3.7).

3.2 **Types and sizes.**

3.2.1 *Type I.* The clips shall be of the type (pattern) and of the sizes (dimensions) shown in figure 1, as specified (see 1.2.1).

3.2.2 *Type II.* The clips shall be of the type (pattern) and of the sizes (dimensions) shown in figure 2, as specified (see 1.2.1).

3.3 **Formation.** Clips shall be formed so that all portions of the wire are approximately parallel and form flat loops, except that there may be a small inward bend near the ends of the wire in type II clips.

3.4 **Damage to paper (type I only).** Gem-pattern clips shall be free from burrs or rough edges. The clips shall meet the test requirements of 4.4.1 without tearing of the paper caused by burrs or rough edges when the item is clipped onto the paper(s) or removed therefrom as described in the test procedure of 4.4.1.

3.5 **Resistance to discoloration.** There shall be no evidence of discoloration of the filter paper when clips have been subjected to the test procedure of 4.4.2 (see 3.7).

3.6 **Diameter and strength of wire.** Clips shall be of the diameter of wire specified in table I, and shall meet the deflection for strength value indicated therein for the respective types and sizes when tested as specified in 4.4.3.

TABLE I.

Type	Size	Diameter of wire	Deflection for strength value
		Inch	*Inch*
I (Gem pattern)	1 (large)	0.041 ± 0.001	0.24
	2 (small)	.036 ± .001	.17
II (Ideal pattern)	1 (large)	.071 ± .002	.75
	2 (medium)	.062 ± .001	.48

3.7 **Workmanship.** The workmanship shall be in accordance with standard commercial practice. The coating specified in 3.1 shall be uniformly smooth and adherent and shall completely cover the clips except the cut ends which may be uncoated (see 3.5).

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier's responsibility.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified hereinafter. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. (See 6.1.)

4.2 **Lot.** All clips of the same type and size, offered for delivery at one time, shall be considered a lot for purposes of sampling for examinations and tests.

4.3 **Sampling and examinations.**

4.3.1 *Sampling.*

4.3.1.1 *Sampling for visual examination.* The sample unit shall be one clip. Not more

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than 10 clips shall be taken from any one box for examination specified in 4.3.2.2. The clips shall be examined visually with a lens of 3X magnification to determine compliance with the coating or finish requirements of 3.7. The sample size shall be in accordance with inspection level S-4 of MIL-STD-105; acceptable quality level (AQL), 4.0 percent defective.

4.3.1.2 *Sampling for dimensional defects.* Sample clips for dimensional examination (see table III) shall be taken at random from the clips selected for examination as specified in 4.3.1.1. The sample size shall be in accordance with inspection level S-2 of MIL-STD-105; AQL, 4.0 percent defective.

4.3.1.3 *Sampling for tests.* Clips, sampled and examined in accordance with 4.3.1.2 and 4.3.2.2, respectively, shall be tested as described in 4.4. The sample size shall be as indicated in 4.3.1.2. The AQL shall be as specified in 4.3.1.2, except for the deflection-for-strength-value test specified hereinafter (see 4.4.3).

4.3.2 *Examinations.*

4.3.2.1 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with table II. For examination of interior packages, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed in table II shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. Not less than five intermediate packages shall be taken at random from each shipping container, and five unit packages from each of the five intermediate packages so taken. Sample packages, taken as specified, shall then be examined for the defects, as applicable, listed in table II. The sample size of shipping containers shall be in accordance with inspection level S-2; AQL, 4.0 defects per hundred units.

TABLE II. *Examination of preparation for delivery*

Examine	Major defects
Marking (interior packages and exterior shipping containers).	Incorrect, incomplete, illegible, omitted; of improper size, location, sequence, or method of application.
Materials (containers).	Component missing, damaged, or otherwise defective affecting serviceability.
Workmanship.	Incomplete closure of case liners, container flaps; loose or inadequate sealing, strapping, or stapling; bulged or distorted containers.
Contents and weight, exterior and interior (as applicable).	Number of packages per shipping container is less than required. Gross or net weight exceeds limitation.
Packaging and packing (interior packages and exterior shipping containers).	Not as specified in section 5. Alternate intermediate packages (5.1.2.1.1) do not retain unit boxes of clips satisfactorily

4.3.2.2 *Examination of end item.* Sample clips, selected in accordance with 4.3.1.1 or 4.3.1.2 shall be examined for the defects, as applicable, listed in table III.

TABLE III. *Classification of defects (end item)*

Defects (major)
Not material required.
Not type and size of item specified.
Diameter of wire not as specified (and within tolerances permitted).
Not in conformance with pattern (design) indicated in figures 1 and 2, as applicable.
Formation of item not as required.
Workmanship not as specified (see 3.7 and 4.3.1.1).

4.4 Tests.

4.4.1 *Damage to paper (type I only).* The Gem-pattern sample clips shall be tested in accordance with the test procedure de-

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scribed below. Size I clips shall be tested with 1, 10, and 25 sheets of paper at a time; size 2 clips with 1, 5, and 10 sheets at a time. Paper used in conducting the tests shall be white writing paper, 32-pound (17 x 22—1,000 sheets), type V, conforming to UU-P-121.

4.4.1.1 *Procedure.* The sample item shall be clipped over the edge(s) of the paper(s), in the manner for which it was designed, care being taken that the movement and position of the clip is parallel to its length and perpendicular to the edge(s) of the paper(s). Each clip shall be tested twice. The opposite side of the inner loop to that which touches the paper in the first test shall be in contact with the paper in the second test. The sample item shall be clipped over the sheet(s) as far as possible without wrinkling the paper(s), and then removed. The clip shall be removed in each instance by placing the thumb on the loop on one side of the paper(s) and the forefinger on the loop on the other side of the paper(s), applying pressure equally with both fingers to slide the clip straight from the paper(s) without turning. Examine paper(s) in each instance for conformance with 3.4. Evidence of tearing of the paper(s) shall be considered as due to defective clip(s).

4.4.2 *Resistance to discoloration.* Paper used in conducting the following test shall be white, Whatman No. 50 hardened filter paper, 7 cm. diameter. Sample clips shall be clipped over the edge of the filter paper in a manner that the cut ends of the wire are not touching the paper. The paper shall then be thoroughly moistened with distilled water and placed in a closed container for a period of 5 hours. The temperature of the container shall be maintained at 80° ± 5° F. during the test. After the specified period of time, withdraw the paper and clips from the container and examine the filter paper for compliance with 3.5. Evidence of any discoloration of the filter paper where sample items were clipped shall be considered as due to defective clip(s). (See 3.7.)

4.4.3 *Deflection for strength value.* Sample clips which have not been subjected to any test shall be tested as follows for the strength requirements of 3.6. When 25 percent or more failure of samples to pass the test of 4.4.3.1 is indicated, the lot shall be rejected. More than 1 failure in 10 to meet the test of 4.4.3.2 shall be cause for rejection.

4.4.3.1 *Type I.* The Gem-pattern clips shall be tested as follows: Place the inner loop of the clip on the end of a firmly supported steel rule and deform the clip by pressing on the other two loops, as shown in figure 3. In applying the load the outside free end of the wire shall be kept perpendicular to the length of the rule. Slowly increase the load until the deflection of the free end indicated by D in figure 3 reaches the value shown in table I. Determination of the deflection shall be accomplished by clamping a slotted plate across the full width of the rule perpendicular to the rule, at the required distance from the end as shown in table I. The clip shall be considered to have failed test if, after the load has been released the permanent deformation of the end of the inner loop is greater than the diameter of the wire. This may be judged by observing the clip against a light background with the line of sight in the plane of the outer loop; excessive deformation will be indicated if it is possible to see light between the inner loop and the outer loop.

4.4.3.2 *Type II.* The Ideal-pattern clips shall be tested as follows: Place the end of a firmly supported steel rule through one loop and against the wires of the other loop. Deform the clip by applying pressure at the ends of the lower loop as shown in figure 4. The portion of the wire on which the load is applied shall be kept perpendicular to the length of the rule and in contact with it. Slowly increase the load until the deflection of the loaded wire indicated by D in figure 4 reaches the value listed in table I. Determination of the deflection shall be accomplished by clamping a slotted plate across

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the full width of the rule, perpendicular to the rule at the required distance from the end, as shown in table I, plus the specified diameter of the wire. The clip shall be considered to have failed the test if after the load has been released the wires of the upper and lower loops do not touch at any one of the "four" points where they cross. This may be judged by placing a single 3- by 5-inch sheet of paper between the loops; excessive deformation will be indicated if the clip does not support the weight of the paper in a vertical position.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and application of the preservation, packaging, and packing levels shall be as defined in Fed. Std. No. 102.

5.1 Packaging.

5.1.1 *Unit packages.*

5.1.1.1 *Levels A and B.*

5.1.1.1.1 *Type I.* Type I, Gem-pattern clips shall be packaged 100 per box, one size only, in a folding paperboard box of appropriate size and conforming to style III, type A, class b of PPP-B-566. Unless otherwise specified, boxes shall have a pyelock tuck-type closure.

5.1.1.1.2 *Type II.* Type II, Ideal-pattern clips shall be packaged 12 per box (size 1, large) or 50 per box (size 2, medium), in boxes conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.1.2 *Level C (types I and II).* Unless otherwise specified, clips shall be unit-packaged in accordance with normal commercial practice.

5.1.2 *Intermediate packages.*

5.1.2.1 *Levels A and B.* (See 6.1.)

5.1.2.1.1 *Type I.* Unit boxes of Gem-pattern clips of one size only (size 1, large; or size 2, small), in quantities as specified in table IV, shall be packaged in folding or set-up paperboard boxes conforming to PPP-B-566 and PPP-B-676, respectively. Alternatively, the unit-packaged clips, as specified in 5.1.1.1, may be intermediate-packaged, in quantities as specified in table IV, in a paperboard sleeve provided with self-locking tabs or tucks. The tabs or tucks shall be arranged and scored in a manner to retain the packaged clips within the sleeve during handling and shipment.

5.1.2.1.2 *Type II.* Twelve unit boxes of size 1 (large) or ten unit boxes of size 2 (medium) clips shall be packaged in folding or set-up paperboard boxes conforming to PPP-B-566 and PPP-B-676, respectively.

5.1.2.2 *Level C (types I and II).* Unless otherwise specified, clips shall be intermediate-packaged in accordance with normal commercial practice.

5.2 Packing. (See 6.1.)

5.2.1 *Level A.* Clips of one type and size, packaged in unit and intermediate containers as specified in 5.1, shall be packed in fiberboard boxes conforming to PPP-B-636, V3c, in quantities as specified in table IV. Boxes shall be closed and sealed in accordance with the appendix to the box specification. Shipping containers shall be provided with waterproof case liners conforming to class 2 of MIL-L-10547.

5.2.2 *Level B.* Clips of one type and size, packaged in unit and intermediate boxes as specified in 5.1.1.1 (level B) and 5.1.2.1 (level B), shall be packed in shipping containers conforming to the requirements of PPP-B-636, type CF or SF, class—domestic, style RSC, in quantities as specified in table IV for the respective types and sizes of clips. The bursting strength of shipping containers shall be a minimum of 200 p.s.i. Closure shall be in accordance with the appendix to the box specification.

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TABLE IV. *Intermediate packaging and packing*

Clips		Unit boxes per intermediate container	Intermediate boxes per shipping container	Gross weight (approx.)
Type	*Size*	*Quantity*	*Quantity*	*Pound*
I	1 (large)	10	25	39
I	2 (small)	10	25	23
II	1 (large)	12	25	65
II	2 (medium)	10	10	48

5.2.3 *Level C.* Clips, packaged as specified in 5.1, shall be packed, unless otherwise specified, in accordance with standard commercial practice in a manner to afford protection against damage during direct shipment from supply service to the first receiving activity for immediate use. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.*

5.3.1.1 *Unit and intermediate packages.* Unit boxes shall be marked with the name of the product, the type, size, and quantity of clips and the name of the manufacturer. Intermediate containers shall be marked as for unit boxes except that they shall be additionally marked with the applicable Federal Stock Number and the quantity of unit boxes contained.

5.3.1.2 *Shipping containers.* In addition to any special marking required in the invitation for bids, contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123, including specification classification (type, size, etc.).

5.3.2 *Military agencies.*

5.3.2.1 *Unit and intermediate packages.* Unless otherwise specified, unit and intermediate packages shall be marked as specified in 5.3.1.1.

5.3.2.2 *Shipping containers.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.4 **Standard pack (civil agencies).** The standard pack requirements for the commodity covered by this specification shall be as specified in 5.1.1.1 (level B), 5.1.2.1 (level B), 5.2.2, and 5.3.1 (see 6.1.1).

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein (see 5.1.1.1.1, 5.1.1.2, 5.1.2.2, 5.2.3, and 5.3.2.1) and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and size of clip required (see 1.2.1).

(*c*) Inspection responsibility if other than as specified herein (see 4.1).

(*d*) Packaging and packing levels and any special marking for shipment required (see 5.1, 5.2, and 5.3).

6.1.1 *Standard pack (civil agencies).* The standard pack requirements in 5.1.1.1 (level B), 5.1.2.1 (level B), 5.2.2, and 5.3.1 are intended for use in procurement of stores stock replenishments. Procuring officers should use the standard pack provisions when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue or shipment to eventual user.

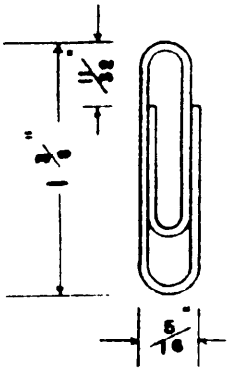
6.2 **Military standards.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

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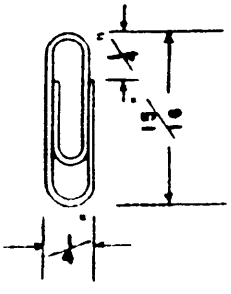
6.3 **Transportation data.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Clips, paper.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.



Type I, Gem Pattern
Size 1, Large

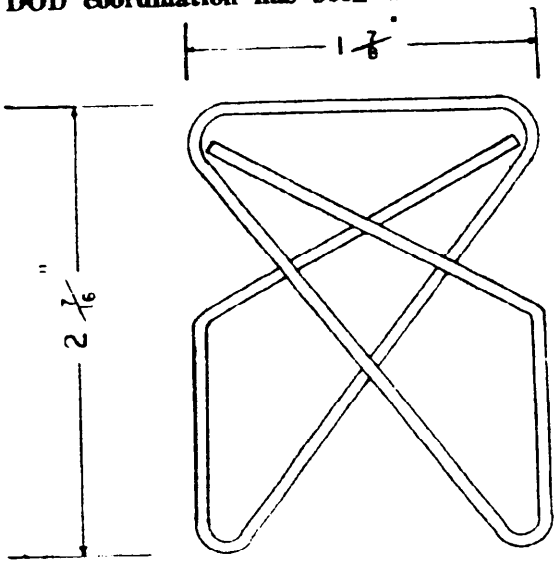


Type I, Gem Pattern
Size 2, Small
Tolerances on dimensions less than 3/8" ± 1/32 in.
Tolerances on dimensions greater than 3/8" ± 1/16 in.
FIGURE 1.

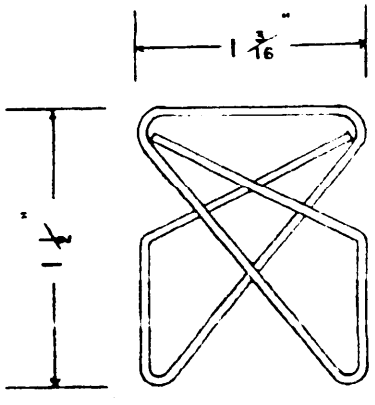
Motor:

Clips, paper.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

DOD coordination has been waived.



Type II, Ideal Pattern
Size 1, Large



Type II, Ideal Pattern
Size 2, Medium
Tolerances on all dimensions ± 1/16 in.
FIGURE 2.

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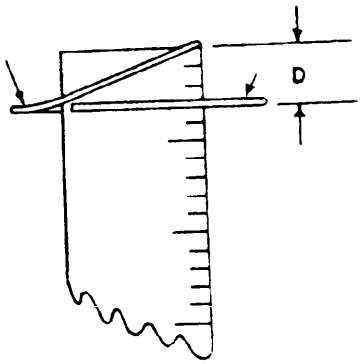


FIGURE 3.

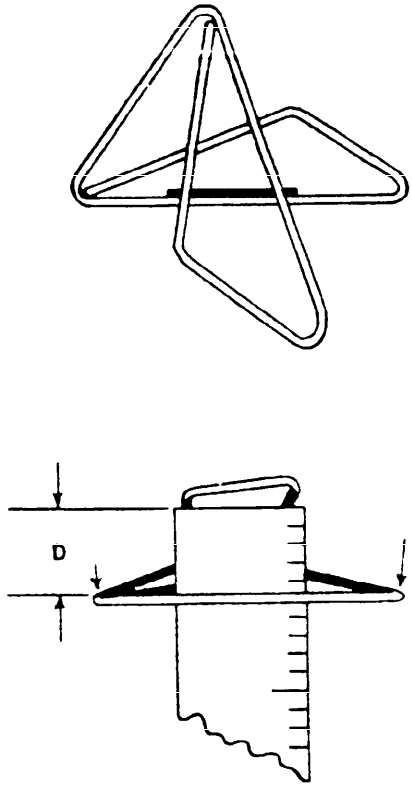


FIGURE 4.

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